



Awesense Energy Transition Platform Use Cases



- demoable implementation



- demoable & open sourced implementation



UC01 Coincident Peak and Non-Coincident Peak Analysis	⦿		
UC02 Power Factor Analysis			●
UC03 Phase Balance Analysis			●
UC04 Transformer Capacity Analysis for EV Charging	⦿		
UC05 Reversed Power Flow and PV Capacity Analysis	⦿		
UC06 GIS Grid Connectivity Validation and Correction			●
UC07 Asset Management - Cables and Lines			
UC08 Asset Management - Substations			
UC09 EV Charging & Use of Reserved Capacity Analysis			
UC10 Connected Loads [kW] vs. Actual Phase Balance Analysis			
UC11 Analysis of Unserved Energy Due to Outage Analysis			
UC12 Grid Temperature Monitoring			
UC13 Master Circuit Breaker Value vs. Measured Values Analysis	⦿		
UC14 Onsite Generation and Ghost Load Analysis			
UC15 Utility Data Challenge Hub			●
UC16 Registered PV Generation Analysis	⦿		
UC17 Actual vs. Contracted Grid Capacity Analysis	⦿		
UC18 Grid Alerts & Warnings			●
UC19 IoT Health Index and Overview			●
UC20 Unexpected Generation Identification			
UC21 Geospatial Analysis of Grid Alerts & Warnings			●
UC22 Topological Grid Segmentation			●
UC23 Analysis of Grid's Segments and Elements			●



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jupyter



Power BI



TGI

UC24 Analysis of Planned Outage for Assets Upgrades			
UC25 Analysis for Minimizing Quantities of IoT and Meter Devices			
UC26 Transformer Loading Analysis			
UC27 IoT Sensor Billing			
UC28 Energy Balancing for Revenue Protection			
UC29 Non-Technical Loss Investigations Management			
UC30 Suspicious Pattern Identification & Analysis			
UC31 Grid and Time Series Visualization			
UC32 Power Outage Visualization			
UC33 MV Grid Optimization Analysis			
UC34 EV & Appliance Disaggregation			
UC35 Standard Consumption Patterns Creation			
UC36 Phase Identification Analysis			
UC37 Detection of Meter Incorrect Wiring Connection			
UC38 Transformer Load Increase Assessment Tool			
UC39 IPT and ICT Verification for Proper Billing			
UC40 Phase Imbalance Correction Suggester			
UC41 EV Load Growth Forecasting			
UC42 Meter to Transformer Association Analysis			
UC43 Load Forecasting			
UC44 Voltage Power Quality Monitoring			
UC45 Feeder Voltage Analysis and Visualisation			
UC46 Feeder Capacity and Lateral Capacity Analysis			



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UC47 Dynamic Line Rating Analysis

UC48 AMI-Smart Meter Flag Analysis for Theft Detection





Awesense Energy Transition Platform Use Cases - Focus Areas

	Consumer Rates & Programms	Situational Awareness and GridOps	Asset Management	Grid Planning	Data Quality Improvement	Billing	Outages	Distributed Energy Resources (DERs)	Multipurpose Data Analytics	Worker Safety	Electric Vehicles (EV)	Non-Wires Alternatives (NWA)	Revenue Protection
UC01 Coincident Peak and Non-Coincident Peak Analysis	✓		✓								✓		
UC02 Power Factor Analysis		✓											
UC03 Phase Balance Analysis		✓											
UC04 Transformer Capacity Analysis for EV Charging			✓	✓							✓		
UC05 Reversed Power Flow and PV Capacity Analysis				✓			✓						
UC06 GIS Grid Connectivity Validation and Correction				✓									
UC07 Asset Management - Cables and Lines			✓										
UC08 Asset Management - Substations			✓										
UC09 EV Charging & Use of Reserved Capacity Analysis					✓						✓		
UC10 Connected Loads [kW] vs. Actual Phase Balance Analysis				✓									
UC11 Analysis of Unserved Energy Due to Outage Analysis							✓						
UC12 Grid Temperature Monitoring		✓	✓										
UC13 Master Circuit Breaker Value vs. Measured Values Analysis	✓			✓		✓						✓	
UC14 Onsite Generation and Ghost Load Analysis							✓						
UC15 Utility Data Challenge Hub								✓					
UC16 Registered PV Generation Analysis							✓						
UC17 Actual vs. Contracted Grid Capacity Analysis				✓								✓	
UC18 Grid Alerts & Warnings		✓											
UC19 IoT Health Index and Overview		✓	✓										
UC20 Unexpected Generation Identification	✓				✓	✓			✓				

[illegible]